

Week 36: Long Run Behavior of Markov Chains (Part 1)

Interested in exploring the long-term behavior of Markov chains

$$\lim_{t \rightarrow \infty} P_{i,j}^{(t)} = \lim_{t \rightarrow \infty} P(X_t = j \mid X_0 = i).$$

For this we need to introduce some additional concepts.

Week 36 (Calculations/Finite state spaces):

- Limiting distribution
- Regular Markov chains
- Stationary distribution
- Interpretation as long-run fraction
- PageRank application

Week 37 (Conceptual/Infinite state spaces):

- Irreducibility
- Recurrence and transience
- Periodicity
- More general limit theorems
- MCMC application?

Week 36

Definition 1.1. (*Limiting distribution*) Consider a DT-MC $\{X_t\}$. We call $\pi = (\pi_0, \pi_1, \dots)^\top$ the *limiting distribution* if the following two conditions are satisfied:

1. The limits $\pi_j = \lim_{t \rightarrow \infty} P_{i,j}^{(t)}$ exist for $j = 0, 1, \dots$ and do not depend on i .
2. $\sum_{j=0}^{\infty} \pi_j = 1$.

Notes:

- 1. $\implies$ 2. for finite state spaces $\{0, 1, \dots, N\}$, but not necessarily for infinite state spaces.
- π_j can be interpreted as the probability of being in state j after many transitions.

Example:

$$\mathbf{P} = \begin{bmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{bmatrix}, \quad \mathbf{P}^{10} \approx \begin{bmatrix} 0.600391 & 0.399609 \\ 0.599414 & 0.400586 \end{bmatrix}, \quad \mathbf{P}^{100} \approx \begin{bmatrix} 0.6 & 0.4 \\ 0.6 & 0.4 \end{bmatrix}$$

Definition 1.2. (*Regular Markov chain*) Consider a DT-MC $\{X_t\}$ with finite state space $\{0, 1, \dots, N\}$ and transition probability matrix $\mathbf{P}$. If there exists a positive integer $k > 0$ so that all elements of $\mathbf{P}^k$ are strictly positive, we call $\mathbf{P}$ and $\{X_t\}$ *regular*.

Problem 1

Are

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{and/or} \quad \mathbf{C} = \begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 0 & 1/2 \\ 0 & 1/2 & 1/2 \end{bmatrix}$$

regular?

Theorem 1.3. *Let $\{X_t\}$ be a regular DT-MC with finite state space $\{0, 1, \dots, N\}$ and transition probability matrix $\mathbf{P}$. Then there exists a unique limiting distribution $\boldsymbol{\pi}$, which is the non-negative solution of*

$$\begin{cases} \boldsymbol{\pi} = \mathbf{P}^\top \boldsymbol{\pi} \\ \mathbf{1}^\top \boldsymbol{\pi} = 1 \end{cases}$$

Note:

- Regularity is a sufficient condition for the existence of a unique limiting distribution.

Definition 1.4. (*Stationary distribution*) A probability distribution $\boldsymbol{\pi} = (\pi_0, \pi_1, \dots)^\top$ is called a *stationary distribution* if

$$\pi_j = \sum_{i=0}^{\infty} \pi_i P_{i,j} \quad \text{for } j = 0, 1, \dots \quad \text{and} \quad \sum_{j=0}^{\infty} \pi_j = 1$$

or, equivalently, $\boldsymbol{\pi} = \mathbf{P}^\top \boldsymbol{\pi}$ and $\mathbf{1}^\top \boldsymbol{\pi} = 1$ for finite state spaces.

Notes:

- If $X_0 \sim \boldsymbol{\pi}$, then $X_t \sim \boldsymbol{\pi}$ for every $t \geq 0$.
- Limiting distribution $\implies$ Stationary distribution (but not the other way around).

Problem 2: Fixing the LF from Exercise 3

Consider the transition probability matrix

$$\mathbf{P} = \begin{bmatrix} p & q & 0 \\ 0 & p & q \\ q & 0 & p \end{bmatrix},$$

where $p, q \geq 0$ and $p + q = 1$. Consider the cases

- $0 < p < 1$ and $q = 1 - p > 0$
- $p = 1$ and $q = 0$.
- $p = 0$ and $q = 1$

For each case, determine whether the Markov chain is regular, does a limiting distribution exist, and/or whether a stationary distribution exists.

Find the $\boldsymbol{\pi} = (\pi_0, \pi_1, \pi_2)^\top$ as a function of p and q .

Definition 1.5. (*Doubly stochastic*) The transition probability matrix $\mathbf{P}$ is called *doubly stochastic* if

$$\sum_{k=0}^N P_{i,k} = \sum_{k=0}^N P_{k,j} = 1$$

for all states i and j .

Theorem 1.6. *Let the Markov chain $\{X_t : t = 0, 1, \dots\}$ be regular with finite state space $\{0, 1, \dots, N\}$. If the transition probability matrix $\mathbf{P}$ is doubly stochastic, then the limiting distribution is uniform*

$$\pi = \left(\frac{1}{N+1}, \frac{1}{N+1}, \dots, \frac{1}{N+1} \right)^\top.$$

Theorem 1.7. (*Long-run fraction of time*) In a regular Markov chain $\{X_t : t = 0, 1, \dots\}$, the limiting distribution $\boldsymbol{\pi} = (\pi_0, \pi_1, \dots, \pi_N)^\top$ gives the long-run fraction of time spent in each state. I.e.,

$$\pi_j = \lim_{n \rightarrow \infty} \mathbb{E} \left[\frac{1}{n} \sum_{k=0}^{n-1} \mathbb{1}\{X_k = j\} \mid X_0 = i \right]$$

for any state i

Problem 3: Putting things together to make our life easy

A Markov chain $\{X_t : t = 0, 1, \dots\}$ has transition probability matrix

$$\mathbf{P} = \begin{bmatrix} 1/2 & 1/2 & 0 & 0 \\ 1/2 & 0 & 1/2 & 0 \\ 0 & 1/2 & 0 & 1/2 \\ 0 & 0 & 1/2 & 1/2 \end{bmatrix}.$$

What is its limiting distribution? What fraction of time, in the long run, does the chain spend in state 0?

Problem 4: Exam August 2025

On any given day Maria is either cheerful (C), so-so (S), or glum (G). If she is cheerful today, then she will be C, S, or G tomorrow with respective probabilities 0.5, 0.4, 0.1. If Maria is feeling so-so today, then she will be C, S, or G tomorrow with probabilities 0.3, 0.4, 0.3. If she is glum today, then she will be C, S, or G tomorrow with probabilities 0.2, 0.3, 0.5. Let X_n denote Maria's mood on day n .

1. • Explain that Maria's mood is a three-state Markov chain.
 - Illustrate with a transition probability graph, and determine the transition probability matrix $\mathbf{P}$.
2. Assume that Maria's mood at day 27 has a uniform probability distribution. What is the probability that Maria is glum at day 28.
3. • Explain that the Markov chain has a unique stationary distribution.
 - How often, on average, is Maria glum?
 - Is it possible that the distribution of X_1 equals the distribution of X_n for all n ?

PageRank



See video

The Strange Math That Predicts (Almost) Anything

Hyperlink-transition matrix

For n pages, let d_i denote the number of links on page i .

$$P_{i,j} = \begin{cases} 1/d_i & \text{if page } i \text{ links to page } j \\ 0 & \text{otherwise.} \end{cases}$$

Example:

$$\mathbf{P} = \begin{bmatrix} 1/2 & 1/2 & 0 & 0 \\ 1/3 & 0 & 1/3 & 1/3 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}.$$

Handling *dangling* pages

If a page has no outgoing links, we replace it by a uniform distribution

$$S_{i,j} = \begin{cases} P_{i,j} & \text{if } d_i > 0 \\ 1/n & \text{if } d_i = 0. \end{cases}$$

Example:

$$\mathbf{S} = \begin{bmatrix} 1/2 & 1/2 & 0 & 0 \\ 1/3 & 0 & 1/3 & 1/3 \\ 1/4 & 1/4 & 1/4 & 1/4 \\ 0 & 1 & 0 & 0 \end{bmatrix}.$$

Ensuring regularity

No guarantee that $\mathbf{S}$ is regular, which we handle through *teleportation*. Let $\alpha \in (0, 1)$ be the probability of following a link, and let $\mathbf{v}$ be a probability distribution with $v_j > 0$.

$$G_{i,j} = \alpha S_{i,j} + (1 - \alpha)v_j.$$

Usually $\alpha = 0.85$ and $\mathbf{v} = (1/n, \dots, 1/n)^\top$.

Example:

$$\mathbf{G} = \alpha \mathbf{S} + \frac{1 - \alpha}{4} \mathbf{1}\mathbf{1}^\top = \begin{bmatrix} 0.463 & 0.463 & 0.038 & 0.038 \\ 0.321 & 0.038 & 0.321 & 0.321 \\ 0.250 & 0.250 & 0.250 & 0.250 \\ 0.038 & 0.888 & 0.038 & 0.038 \end{bmatrix}.$$

Finding the limiting distribution

From regularity, we know that the limiting distribution π exists and is unique. It satisfies $\pi = \mathbf{G}^\top \pi$ and $\mathbf{1}^\top \pi = 1$.

To find it quickly, we can start with an initial distribution $\pi^0 = (1/n, \dots, 1/n)^\top$ and iterate

$$\pi^{k+1} = \mathbf{G}^\top \pi^k$$

until convergence. This is called the *power method*.

Implementation in Python